

Pengyu Zhang

✉ p.zhang@uva.nl 🌐 pengyu-zhang.github.io 🔗 LinkedIn 🎓 Google Scholar

📍 Amsterdam, The Netherlands

Summary

Computer Science Ph.D. candidate (UvA) seeking Data Scientist roles. I build deep learning models for multi-modal knowledge graphs that stay accurate as data drifts over time and that tell near-identical entities apart, fusing graph structure, text, images, and temporal signals for entity linking, link prediction, and recommendation. My methods consistently beat prior state-of-the-art (up to +**20%** on entity linking, +**6%** on recommendation) and ship as open-source code with released datasets and benchmarks. Eight first-author papers across these directions, with venues including ESWC, CIKM, and ECAI, and two under review at KDD and PKDD.

Skills

Programming Python, SQL, Bash, Git, Linux.

ML & DL PyTorch, Hugging Face, scikit-learn, NumPy, Pandas; large language models, vision-language models, graph neural networks, contrastive learning, recommendation.

Tools Weights & Biases, Jupyter.

Research & Project Experience

TimeRoute **Time-aware multi-modal recommendation** *Under review, KDD 2026*

- Lifted recommendation accuracy by up to **6%** on TikTok, Amazon-Baby, and Amazon-Sports over strong baselines.
- Recommenders typically blend user signals (clicks, text, images) the same way regardless of when they happened. Built a model that learns which signals matter over short versus long time spans and weighs them accordingly, then automatically cleans up noisy and missing interactions so recommendations stay reliable.
- Stack: PyTorch, graph neural networks, diffusion models; tracked with W&B.

Time Imprint **Time as a modality for entity disambiguation** *Under review, PKDD 2026*

- Boosted top-match accuracy by up to **4.81%** overall and up to **200%** on the hardest, most confusable cases.
- Systems often confuse near-identical records whose text and images look almost the same. Added timing as an extra signal so the model can tell them apart, sharply cutting errors on the most confusable pairs while keeping the rest accurate.
- Stack: PyTorch, contrastive learning, multi-modal fusion.

Beyond Images **LLM/VLM data-enrichment pipeline for knowledge graphs** *ESWC 2026*

- Boosted match accuracy by up to +**7%** overall and +**333%** on ambiguous logo and symbol subsets.
- Many records have missing or low-quality images, which hurts matching. Built an automated pipeline that finds extra images online, turns them into text with vision-language models, and summarizes everything with an LLM, filling the gaps without any manual work.
- Released open-source code and reproducible benchmark. 🔄 Code

CYCLE **Contrastive learning for temporally robust entity linking** *CIKM 2024*

- Beat the best prior method by **13.9%** (1-year gap) and **17.8%** (3-year gap), with the largest gains on rare records.
- Models that match text to a database get worse as that database changes from year to year. Designed a training approach that learns from those yearly changes so accuracy stays high as the data evolves, and released an open benchmark (GCL-TempEL, built on Wikidata5M) for the problem.

- Released open-source code. 🔄 Code

TIGER **Graph-enhanced entity linking under temporal drift** *ECAI 2024*

- Outperformed the strongest baseline by +**16.24%** (1-year gap) and +**20.93%** (3-year gap), with strong gains on well-connected records.
- Tackling the same drift problem, combined how records connect to each other with their text descriptions to make matching more robust as data changes over time, and released a public benchmark (Graph-TempEL) to measure it.
- Released open-source code. 🔄 Code

Graph-TempCZ

Large-scale graph for software-usage prediction

LREC 2026

- Improved test accuracy by **5.98%** (to **92.88%**) with a GraphSAGE GNN over feature-based XGBoost baselines on a link-prediction task.
- Built the first large-scale graph linking publications to the software they use (over six million mention relationships, 1959 to 2022) and studied how well predictions hold up across years.
- Stack: PyTorch Geometric, GraphSAGE, XGBoost, Sentence-BERT; large-scale data engineering.

Fusion Training

Training dynamics of hybrid-reasoning LLMs

ACL 2026 SRW

- Newer LLMs switch between quick answers and long step-by-step reasoning to save compute, but training both behaviors into one model makes them compete. Showed that interleaving the two kinds of training data keeps both skills strong, and quantified the trade-off (more non-thinking data weakens reasoning, $r = -0.58$).
- Ran a full factorial study over data ratios and training schedules, and released an open benchmark (Fusion Bench) for the community.
- Stack: Qwen3-4B, LoRA, Hugging Face Transformers/TRL, W&B; released a benchmark.

Education

2022 – 2026

Ph.D. in Computer Science, University of Amsterdam (UvA), The Netherlands.
Supervisors: Prof. Dr. Paul Groth, Dr. Klim Zaporozets. Focus: multimodal knowledge graphs, temporal entity linking, recommendation.

2019 – 2022

M.Eng. in Control Engineering, Beijing University of Technology (BJUT), China.

Selected Publications

Full list on Google Scholar. (**bold** = first author)

First-author

- **P. Zhang**, Y. Jiang, K. Zaporozets, C. Cao, P. Groth. *TimeRoute: Time-Scale Experts for Multi-Modal Diffusion Recommendation*. Under review, KDD 2026. [🔗](#) Code
- **P. Zhang**, K. Zaporozets, C. Cao, J.-H. Huang, P. Groth. *Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs*. Under review, PKDD 2026. [🔗](#) Code
- **P. Zhang**, K. Zaporozets, J. Liu, J.-H. Huang, P. Groth. *Are a Thousand Words Better Than a Single Picture? Beyond Images: A Framework for Multi-Modal Knowledge Graph Dataset Enrichment*. ESWC 2026. [🔗](#) Code
- **P. Zhang**, C. Cao, K. Zaporozets, P. Groth. *CYCLE: Cross-Year Contrastive Learning in Entity-Linking*. CIKM 2024. [🔗](#) DOI [🔗](#) Code
- **P. Zhang**, C. Cao, P. Groth. *TIGER: Temporally Improved Graph Entity Linker*. ECAI 2024. [🔗](#) DOI [🔗](#) Code
- **P. Zhang**, Y. Zhang, X. Piao, Y. Sun, B. Yin. *Relationship Updating Network with Contrastive Learning*. Physica A 2024. [🔗](#) DOI [🔗](#) Code
- **P. Zhang**, Y. Zhang, J. Wang, B. Yin. *MVMA-GCN: Multi-view Multi-layer Attention Graph Convolutional Networks*. EAAI 2023. [🔗](#) DOI [🔗](#) Code
- **P. Zhang**, Y. Zhang, Y. Cui, B. Yin. *Visual Analysis for Name Disambiguation of Academic Papers (in Chinese)*. JCAD 2022. [🔗](#) DOI

Selected collaborations

- M. Li, **P. Zhang**, et al. *A Survey of Large Language Models for Data Challenges in Graphs*. Expert Systems with Applications (ESWA) 2025. [🔗](#) DOI [🔗](#) Code
- C. Cao, **P. Zhang**, J. Bloem. *Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications*. LREC 2026. [🔗](#) DOI
- C. Cao, **P. Zhang**, J. Bloem. *Fusion Training for Mathematical Generalization in Large Language Models*. ACL 2026 (Student Research Workshop). [🔗](#) Link

Languages

Chinese (native); English (fluent, IELTS 7.0); Dutch (elementary, A2).